

Separable Equations



Order and verifying solutions

- 1 State the order of the differential equation $y'' + 3y' - 2y = \sin x$.
 - 2 Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.
-

Recognizing and solving separable equations

- 3 Solve $\frac{dy}{dx} = xy$.
 - 4 Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.
-

Initial value problems

- 5 Solve $\frac{dy}{dx} = 3x^2y$, $y(0) = 2$.
 - 6 Solve $\frac{dy}{dx} = -2xy$, $y(0) = 5$.
-

Application: radioactive decay and population growth

- 7 A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k .
 - 8 A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.
-

Further practice: separable equations

- 9 Solve $\frac{dy}{dx} = \frac{1 + y^2}{1 + x^2}$.
 - 10 Solve $y' = y \cos x$, $y(0) = 1$.
-

More separable equations

11 Solve $\frac{dy}{dx} = \frac{x^2}{y}$, $y > 0$.

12 Solve $\frac{dy}{dx} = e^{x-y}$.

A separable equation solved via partial fractions

13 Solve $\frac{dy}{dx} = y(1-y)$ (the logistic equation) using $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.

14 Solve $\frac{dy}{dx} = 2xy^2$, $y(0) = 1$.

Slope fields and qualitative behavior

15 For $\frac{dy}{dx} = x - y$, what does the slope field look like along the line $y = x$?

16 For $\frac{dy}{dx} = y^2$, explain why any solution with $y(0) > 0$ must be increasing for as long as it's defined.

Verifying a solution to an applied model

17 Newton's Law of Cooling states $\frac{dT}{dt} = -k(T - T_{\text{room}})$. Verify that $T(t) = T_{\text{room}} + (T_0 - T_{\text{room}})e^{-kt}$ solves it.

A final applied separable model

18 A cup of coffee at 90°C is placed in a 20°C room. Newton's Law of Cooling gives $T(t) = 20 + 70e^{-kt}$. If the coffee cools to 70°C after 10 minutes, find k .

One more round of separable-equation practice

19 Solve $\frac{dy}{dx} = \frac{x+1}{y}$, $y(1) = 2$.

20 Solve $\frac{dy}{dx} = \frac{y}{x+1}$, $x > -1$.

A final pair of separable equations

21 Solve $\frac{dy}{dx} = \frac{y}{x}, x > 0$.

22 Solve $\frac{dy}{dx} = 3y, y(0) = 4$.